

Can AI chatbots help retain customers? Impact of AI service quality on customer loyalty

Impact of AI
service quality

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Abstract

Purpose – This study investigates whether and how the service quality of artificial intelligence (AI) chatbots affects customer loyalty to an organization.

Design/methodology/approach – Based on the sequential chain model of service quality loyalty, this study first classifies AI chatbot service quality into nine attributes and then develops a research model to explore the internal mechanism of how AI chatbot service quality affects customer loyalty. The analysis of survey data from 459 respondents provided insights into the interrelationships among AI chatbot service quality attributes, perceived value, cognitive and affective trust, satisfaction and customer loyalty.

Findings – The results show that AI chatbot service quality positively affects customer loyalty through perceived value, cognitive trust, affective trust and satisfaction.

Originality/value – This study captures the attributes of the service quality of AI chatbots and reveals the significant influence of service quality on customer loyalty. This study develops research on service quality in the information system (IS) field and extends the sequential chain model of quality loyalty to the context of AI services. The findings not only help an organization find a way to improve customers' perceived value, trust, satisfaction and loyalty but also provide guidance in the development, adoption, and post-adoption stages of AI chatbots.

Keywords AI chatbots, Service quality, Satisfaction, Cognitive trust, Affective trust, Perceived value, Customer loyalty

Paper type Research paper

1. Introduction

Customer loyalty, a perennial concern of both business and academia, reflects a customer's positive attitude toward an organization. Loyal customers believe that, compared with competitors, the focal organization can provide better services and products to them, and they will demonstrate repurchase intention and behavior or retain the services and products of the organization (Harris and Goode, 2004). Hence, customer loyalty is an important factor in an organization's profit increase (Maklan *et al.*, 2017). As artificial intelligence (AI) chatbots may be the first objects that customers contact with organizations, the perceived quality of AI chatbots significantly affects customers' impressions of the organization. Studies show that

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customers' perceived high service quality causes customer satisfaction, which further leads to loyalty (Maklan *et al.*, 2017). This chain model of the impact of service quality on customer loyalty is challenging in the context of AI services. AI is increasingly reshaping the service industry through the entry of various AI-enabled service jobs (Huang and Rust, 2018). A frontline AI chatbot is a typical example of this. The context of frontline customer service has changed with the development of AI chatbots.

An AI chatbot is a conversation system supported by AI technology of natural language understanding, machine learning and big data analysis (Cheng *et al.*, 2021; Lee *et al.*, 2021). AI-embedded systems can make activities more effective and efficient (Liang *et al.*, 2021). Enabled by AI technology, the specific abilities of some AI chatbots surpass those of human workers, such as information acquisition, memory and calculation. As organizations use AI chatbots more extensively and deeply, scholars in the management field are increasingly focusing on frontline AI chatbots. Previous studies have summarized the characteristics and functions of AI chatbots. AI chatbots provide 24/7 services and answer queries effectively, saving costs for organizations (Ashfaq *et al.*, 2020; Chen *et al.*, 2020). They have human-like cues in visual and natural language communication, name, identity (Araujo, 2018), empathy, or friendliness (Cheng *et al.*, 2021). Moreover, AI chatbots will never experience a sense of job burnout or exhaustion. They always responded to customers immediately and energetically. As the services provided by AI chatbots and human beings are different, evaluating service quality with the same standard is unsuitable. Therefore, the relationship between AI chatbot service quality and customer loyalty remains unclear in this new context.

Reviewing the research on AI chatbots, previous studies have explored the evaluation of chatbots and the outcomes of user-AI chatbot interaction. Regarding the evaluation of AI chatbots, Li *et al.* (2021) proposed understandability, reliability, responsiveness, assurance and interactivity to assess AI chatbots in an online context of a travel agency. Ashfaq *et al.* (2020) suggested using perceived enjoyment, perceived usefulness, perceived ease of use, information quality and service quality to evaluate AI chatbots. Other factors used to assess AI chatbots include anthropomorphism (Sheehan *et al.*, 2020), accessibility, flexibility (Ashfaq *et al.*, 2020), empathy, friendliness (Cheng *et al.*, 2021), immediate response and availability (Chen *et al.*, 2020). Regarding the various consequences of AI chatbot use, previous studies have investigated adoption (Sheehan *et al.*, 2020), satisfaction (Crolic *et al.*, 2022), trust (Chen *et al.*, 2020), continued use (Li *et al.*, 2021), purchase (Luo *et al.*, 2019), firm evaluation (Crolic *et al.*, 2022), affective attachment to an AI chatbot (Lee *et al.*, 2021), perceived value (Wang *et al.*, 2021), customer experience and brand love (Trivedi, 2019). Jenneboer *et al.* (2022) summarized the impact of chatbots on customer loyalty. Although existing studies have noticed that service quality may affect loyalty, there is a lack of empirical studies that examine this relationship. Most studies in this review have investigated customer trust, satisfaction and commitment in the context of AI chatbots. Table A1 in Appendix A reviews the research on AI chatbots.

Based on the brief review above, we find two gaps worth studying in the literature. First, existing studies used the constructs of traditional information system attributes to describe AI chatbot service quality, such as assurance, responsiveness, reliability, interactivity, usefulness and ease of use (see Table A1 in Appendix A). These attributes are general and do not reflect the characteristics of AI chatbots. Some studies captured service quality-related attributes. However, this research is somewhat fragmented. Notably, most studies have focused on one or two attributes of AI chatbots, such as anthropomorphism (Crolic *et al.*, 2022), empathy, friendliness (Cheng *et al.*, 2021), decisional guidance and communication style (Chen *et al.*, 2020). Some studies conceptually propose the characteristics of AI chatbots but lack empirical studies, such as Stieglitz *et al.* (2022). AI chatbots serve customers directly on the frontline; hence their service quality is essential to both customers and organizations. De Keyser *et al.* (2019) also called for research evaluating AI chatbots' performance. Second,

although customer loyalty is vital to organizations, few studies have examined the antecedents of customer loyalty in the context of AI services. However, the impact of service quality on customer loyalty remains unclear.

Therefore, we proposed two research questions:

RQ1. What are the attributes of the service quality of AI chatbots?

RQ2. Does service quality affect customer loyalty?

Based on the sequential chain model of service quality loyalty (Harris and Goode, 2004; Semeijn *et al.*, 2005), we developed a structural model to test how AI chatbot service quality affects customer loyalty. This study mainly contributes to theories and practice in the following ways. First, it enriches the research on service quality in the information system (IS) field. We capture the attributes of AI in the context of frontline customer service. As AI chatbots directly provide services to customers on the frontline, we verified that their service quality significantly affects customer loyalty. Second, this study extends the sequential chain model of service quality loyalty to a new context.

On the one hand, as the starting point of the sequential chain model, the service quality of human workers and machines significantly differs. Whether the sequential chain model is still valid for AI services has not yet been studied. This study verifies the effectiveness of this framework in an AI service environment. At the same time, this study extends this sequential chain model of quality loyalty by revealing AI chatbot service quality factors and classifying trust. Third, for managers, our research findings help organizations understand the significant effect of AI chatbot service quality. Moreover, the attributes of AI chatbot service quality that we summarize will guide organizations in the development, adoption and post-adoption stages of AI chatbots.

2. Theoretical background: sequential chain model of service quality-loyalty

The research model in this study was developed based on Harris and Goode's (2004) sequential chain model of quality loyalty. Service quality is an important research topic in this field. Many empirical studies have demonstrated the positive impact of service quality on customer loyalty (Bernardo *et al.*, 2012; Dai *et al.*, 2011; Yuan *et al.*, 2020). In addition, studies have explored the internal mechanism by which service quality influences customer loyalty and found two key mediators: satisfaction and perceived value (Arora and Narula, 2018; Bolton *et al.*, 2000; Rust and Oliver, 2000). After reviewing the literature on service quality and loyalty, Harris and Goode (2004) proposed a sequential chain model of service quality loyalty that integrates trust into the research model. Therefore, they built a sequential chain model, as shown in Figure 1. This sequential chain model has been widely

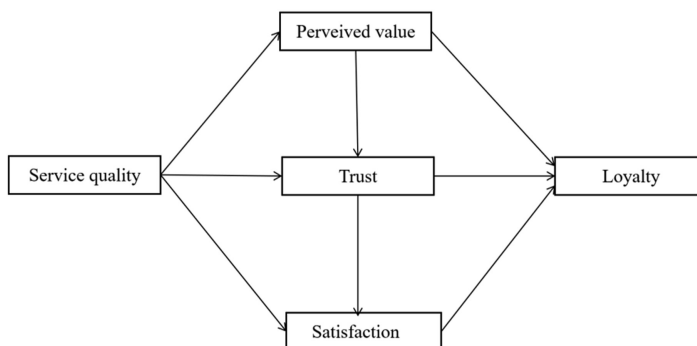


Figure 1.
Sequential chain model
of service quality-
loyalty in Harris and
Goode (2004)

used in loyalty research, both completely and partially. For example, [Aydin and Özer \(2005\)](#) used the service quality–trust–loyalty framework to study customer loyalty in the Turkish mobile telecommunication market. [Chen *et al.* \(2019\)](#) investigated users' recommendation behavior for brand apps by adopting a value-satisfaction-loyalty framework. Using the service quality-satisfaction-loyalty chain, [Yuan *et al.* \(2020\)](#) study the antecedents of mobile payment loyalty.

Satisfaction, trust and perceived value are customers' psychological responses after consuming a service or product ([Oliver, 1997](#)). These three concepts describe customers' different psychological states. Satisfaction is a customer's psychological state and feelings of pleasure or disappointment after comparing the actual performance and expectations of the product or service ([Arora and Narula, 2018](#); [Oliver, 1997](#)). Additionally, satisfaction is a primary predictor of customer behavior ([Lee and Seong, 2020](#)). Meanwhile, perceived value reflects a customer's overall evaluation of a product or service ([Hsiao and Chen, 2016](#)). Customers' perceived high value is a major motivation for purchasing ([Yang and Peterson, 2004](#)). Trust is the willingness to make oneself vulnerable to actions taken by a trusted party based on a feeling of confidence or assurance ([Gefen, 2002](#)). These three types of psychological responses are positively associated with high service quality ([Kuo *et al.*, 2009](#); [Kursunluoglu Yarimoglu, 2017](#); [Liao *et al.*, 2011](#)). Previous studies used satisfaction to evaluate service quality ([Ladhari, 2008](#)). Accordingly, trust is a feeling of assurance brought about by high service quality ([Gefen, 2002](#)). Loyalty is an outcome of underlying positive psychological factors ([Arora and Narula, 2018](#)). Satisfied customers who perceive high-value services and have a trusting attitude tend to have a higher loyalty level than those who do not in these psychological states.

The service context of past studies focuses on traditional offline, e-commerce and m-commerce, neglecting the research on customer loyalty in the context of AI service provision. In this context, direct service providers have changed. As service quality is the starting point of the sequential chain model of loyalty, the validity of the sequential chain model in the AI service context needs to be examined.

Reviewing the literature on service quality, we found that much research has been conducted on the service quality of human service providers. The most notable research findings on service quality are the SERVQUAL dimensions and scale proposed by [Parasuraman *et al.* \(1988\)](#), including tangibles, responsiveness, assurance, reliability and empathy. The five-element framework has been verified in both offline and online contexts such as e-commerce ([Gefen, 2002](#); [Semeijn *et al.*, 2005](#)) and information systems (IS) ([Jiang *et al.*, 2002](#); [Kettinger, 1997](#); [Tate and Evermann, 2010](#)). With the development of information technology, many studies on service quality specific to the online context have emerged. For example, research topics include classifications of service quality in the online purchasing context ([Barnes and Vidgen, 2002](#); [Yoo and Donthu, 2001](#)), service quality of websites ([Barnes and Vidgen, 2002](#); [Cai and Jun, 2003](#)) and service quality of e-services ([Fassnacht and Koese, 2006](#); [Santos, 2003](#); [Tan *et al.*, 2013](#); [Xu *et al.*, 2013](#)). Existing classifications cannot accurately display the characteristics of AI chatbot service quality. We summarize the existing service quality dimensions in [Table A1](#) of Appendix 1.

Based on the literature review of the existing service quality dimensions, they cannot fit well in a specific AI service situation. First, existing service quality dimensions are summarized based on specific and non-AI-related research contexts. Most of the attributes reflect the characteristics of their research context, which are not applicable to AI chatbot service contexts, such as web store policies ([Madu and Madu, 2002](#)), call-back systems ([Li *et al.*, 2002](#)), website design ([Wolfenbarger and Gilly, 2003](#)), replacing ([Cenfetelli *et al.*, 2008](#)) and on-time delivery ([Luo *et al.*, 2012](#)). Second, some dimensions are relatively general, such as efficiency ([Zeithaml *et al.*, 2000](#); [Parasuraman *et al.*, 2005](#)), communication ([Cox and Dale,](#)

2001), service interaction (Barnes and Vidgen, 2002), trustworthiness (Cai and Jun, 2003), fulfillment (Surjadaja *et al.*, 2003) and information quality (Fassnacht and Koese, 2006). These concepts can be applied in various service contexts. Although these attributes are important, we argue that they cannot accurately capture the specific contextual characteristics of the AI chatbot service quality. Third, existing service quality dimensions cannot represent the features of an AI chatbot's service, such as AI's human-like, self-learning and natural language interaction characteristics. Indubitably, exploring the content of AI chatbot service quality is necessary. The nine attributes of service quality in this study reflect the factors that customers are concerned about and the context-specific characteristics, including accuracy of responses, availability of human service alternatives, self-learning, personalized recommendations and human-like empathy.

In this study, we develop a sequential chain model of service quality loyalty in an AI service scenario. In this context, we explore the elements of AI chatbot service quality. When analyzing the elements of AI chatbot service quality, the study finds that these elements may impact customers' cognitive and emotional trust. Therefore, in the hypothesis development, we also discuss the impact of AI chatbot service quality on these two types of trust. The types of trust come from the literature (Johnson and Grayson, 2005), and the factors of AI chatbot service quality come from two sources: recent industry reports and interview data. We recruited 50 college students for semi-structured interviews. In a face-to-face manner, we asked them questions about their perceptions of the quality of the AI chatbot service and recorded their answers for text analysis. Appendix 2 presents a semi-structured interview questionnaire.

We conducted a content analysis of the interview data using a coding technique (Hutchison *et al.*, 2010). The coding process mainly includes three steps. First, we conducted open coding. Two coders performed independent double-blind coding, extracting the keywords of the AI chatbot service quality from the interview materials. Second, after obtaining the open-code results from the two coders, we calculated inter-coder reliability, which reached 0.95. This value indicates that the inter-coder reliability is acceptable (Venkatesh and Brown, 2001; Miles and Huberman, 1994). We then compared the meanings of the open codes if coders have conflicting opinions. We returned to the original sample and discussed it until a consensus was reached. Third, we summarized the open codes as axial codes. The final axial code results show that the characteristics of AI chatbot service quality include the availability of human service alternatives, self-learning, ease of use, personalized recommendations and human-like empathy. We present examples of interview samples, open codes and axial codes in Table A3.

From recent industrial reports, we find that the most prominent characteristics of AI chatbots are that they are always available, omnipresent, provide services with consistent quality and effectively handle customer requirements based on NLP (Bakhasi, 2018; Deloitte, 2018a, 2018b). These characteristics are consistent with customer perceptions of the quality of AI chatbot services. To ensure the mutual exclusivity of the dimensions, after we summarize the nine dimensions of AI chatbot service quality, we return to the literature to compare their definitions (e.g. Venkatesh *et al.*, 2011; Baek and Yoo, 2018; Alge, 2001). We define the constructs contextually by considering both the literature and open codes. We can infer that the dimensions are mutually exclusive based on the different definitions of the constructs. The definitions of the constructs in the research model are presented in Table 1.

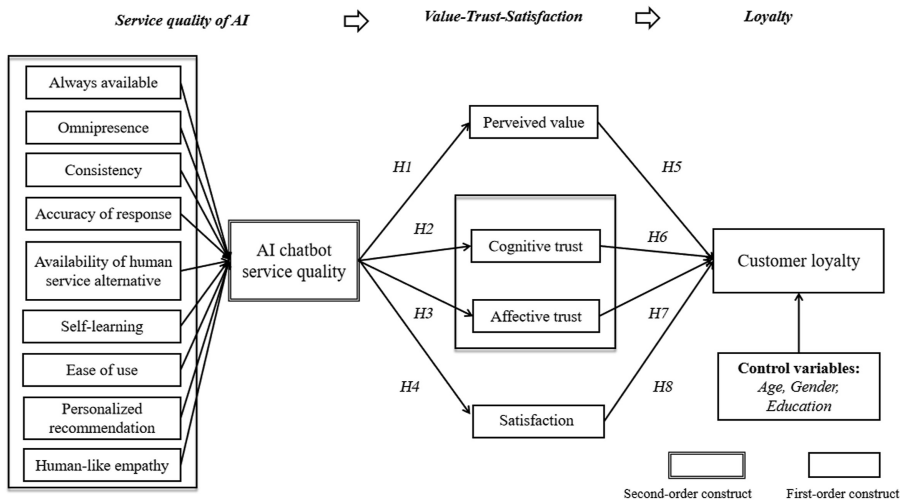
3. Research model and hypothesis

We develop a research model to study how AI chatbot service quality affects customer loyalty to an organization, as shown in Figure 2.

Table 1.
Definitions of the
constructs

Constructs	Definitions	Related literature
Always available	The customers can get AI chatbot's service at any time	Venkatesh <i>et al.</i> (2011)
Omnipresence	Using AI chatbot's service is not limited by place and device	Baek and Yoo (2018)
Consistency	The service quality provided by an AI chatbot is stable	Alge (2001)
Accuracy of Response	AI chatbots can respond to customers' requirements accurately	Mayer and Davis (1999)
Availability of human service alternatives	Human service is readily available when using AI chatbots	Furneaux and Wade (2011)
Self-learning	An AI chatbot continuously improves functions by self-learning	Rijsdijk <i>et al.</i> (2007)
Ease of use	The AI chatbot is easy to use	Gefen <i>et al.</i> (2003)
Personalized recommendation	The AI chatbot recommends items based on customer preferences	Zhang <i>et al.</i> (2011)
Human-like empathy	Customers perceive that AI chatbots can understand them and serve them attentively	Pitt <i>et al.</i> (1995)

Figure 2.
Research model



3.1 AI chatbot service quality and perceived value

Perceived value refers to “the consumer’s overall assessment of the utility of a product or service based on perceptions of what is received and what is given” (Zeithaml, 1988, p. 14). The concept of perceived value involves a trade-off between the perceived “sacrifices given” and “benefits received” (Lin *et al.*, 2012). Perceived benefits usually come from the quality, functionality and features of a product or service (Lindić and Marques da Silva, 2011). Perceived costs include monetary and nonmonetary payments or sacrifices. Nonmonetary sacrifices include users’ perceived cost of time, effort and risk (Lindić and Marques da Silva, 2011). The level of customer perceived value comes from comparing the benefits and costs.

Customers’ perceived service value depends on the service provider’s ability to satisfy their needs (Zehir and Narcikara, 2016). As service quality reflects customers’ overall

subjective judgment of the service (Parasuraman *et al.*, 1988), it affects customers' perceived value. Existing studies empirically validate that high service quality is positively associated with perceived customer value (Kettinger *et al.*, 2009; Thaichon *et al.*, 2014).

In this research context, the nine AI chatbot service quality factors can either increase the perceived benefit or decrease cost. For *always available* and *omnipresence*, customers gain the benefits of accessing services anytime and anywhere while reducing the time and effort spent searching for the service. *Consistency* improves the certainty of service results. *Accuracy of response* ensures smooth communication. An AI chatbot understands customer requirements and provides accurate feedback. *Availability of human service alternatives* effectively avoids service failure when an AI chatbot cannot serve customers well. *Self-learning* makes customers enjoy the benefits of service quality contentious improvement. *Ease of use* decreases customers' learning costs and time spent. *A personalized recommendation* can help customers reduce their search costs when they obtain products that meet their preferences. *Human-like empathy* gives customers emotional value. Hence, we hypothesize as follows:

- H1. The AI chatbot service quality is positively related to the user's perceived value of AI chatbots.

3.2 AI chatbot service quality and trust

Customers begin a rational evaluation process and generate two types of trust in the interaction process: cognitive trust and affective trust. Cognitive trust means an individual determines whether the object is trustworthy based on rational analysis (Chai *et al.*, 2015). The basis of judgment comes from the evidence of the service provider's performance and expertise, such as reliability, competence and credibility (Johnson and Grayson, 2005). In the context of AI chatbot services, customers' perceptions of the trustee's expertise and competence are mainly reflected in four aspects: accuracy of the response. The accuracy of a response is based on the recognition or understanding of AI. Only if the requirements of customers can be understood by AI chatbots can they respond to customers effectively and satisfy their needs (Li *et al.*, 2021). Otherwise, the lack of responsive accuracy leads to needs recognition failure, which is a typical functional failure of online services (Tan *et al.*, 2017). The second competency of AI chatbots reflects the availability of human service alternatives. Currently, AI chatbots can perfectly handle standard and straightforward requirements. However, sometimes customers' questions are uncommon, rare, or complex; therefore, human service intervention is necessary (Ba *et al.*, 2010). On this occasion, the AI chatbot should have the ability to recognize whether the customers need to transfer to human service and respond to customers appropriately to decrease the negative effect of incomplete service. The third is self-learning, a significant difference between AI chatbots and traditional SST technology. Based on big data and machine learning technologies, as AI chatbots deal with more problems, the scope of the problems they can solve will become larger. After a period of AI chatbot use, customers may find that it becomes smarter and service quality improves. The fourth is personalized recommendations. Personalized recommendations are essential for successful customer service (Zhang *et al.*, 2011). By obtaining and analyzing customer online behavior data, AI technology can capture customer preferences in real time and make more accurate personalized recommendations. The above four main AI chatbots' competences in service provide the basis for customers' rational evaluation of whether an AI chatbot is trustworthy. Hence, we hypothesize as follows:

- H2. The AI chatbot service quality is positively associated with customers' cognitive trust toward AI chatbots.

Affective trust is based on customers' moods, emotions, or feelings generated by their perceptions of the service provider's care, concern, patience, friendliness, politeness and

empathy (Johnson and Grayson, 2005). The AI interactive service system was designed to act empathically (Yalcin and Dipaola, 2018). The empathy of frontline employees is always one of the most critical capabilities, and AI chatbots' empathy makes customers feel pleasure as they perceive their emotional state to be understood (Varca, 2009). Meanwhile, AI's ability to provide accurate recommendations not only allows customers to perceive the machines' superior calculation and service abilities but also brings customers the feeling of being understood, generating the sense that the AI chatbot is concerned about them and simultaneously causes cognitive as well as affective trust. Hence, we hypothesize as follows:

- H3.* The AI chatbot service quality is positively associated with customers' affective trust toward AI chatbots.

3.3 AI chatbot service quality and satisfaction

Satisfaction is customers' summary psychological response to their experience of the entire process of using a service or product. Whether a customer feels satisfied depends on the comparison between their expectation and actual experience in the pre- and post-consumption (Arora and Narula, 2018; Oliver, 1997). Prior studies, such as Arora and Narula (2018) and Liang *et al.* (2020), investigated the relationship between satisfaction and service quality. Service quality is an important predictor of customer satisfaction (Deng *et al.*, 2010). As we analyzed in *H1*, while increasing the perceived benefits, AI chatbot service quality reduces perceived costs, including learning, energy and time costs. Customers who perceive a service as having a high value tend to be satisfied (Chen *et al.*, 2019; Xu *et al.*, 2015). Therefore, we hypothesize as follows:

- H4.* The AI chatbot service quality positively affects customers' satisfaction.

3.4 Relationship between perceived value, trust, satisfaction and customer loyalty

Customer loyalty refers to "a deeply held commitment to rebuy or re-patronize a preferred product/service consistently" (Hoehle and Venkatesh, 2015, p. 457). Even in the face of unfavorable information about the organization, the attitudes and behaviors of loyal customers are less affected (Yuan *et al.*, 2020). An AI chatbot directly contacts customers as frontline service providers. We argue that customers' perceptions of and attitudes toward AI chatbots influence their loyalty to the organization in this research context. Customer value is regarded as the basis for organizations' marketing activities (Yang and Peterson, 2004). Many qualitative and quantitative studies have demonstrated the significant effect of perceived value on customer loyalty (Hsiao and Chen, 2016; Hurley and Laitamaki, 1995; Yang and Peterson, 2004).

Trust influences customers' positive attitudes toward organizational behavior and motivation (Hohenstein and Jung, 2020). In the context of AI chatbot services, cognitive trust reflects customers' confidence in AI chatbots' competence (Chai *et al.*, 2015), whereas affective trust represents customers' feelings about the likeability of AI chatbots. Both cognitive and affective trust in AI chatbots cause customers to form a positive attitude toward the organization. Based on the trust transfer theory, customers transfer trust from AI chatbots to the organization (Lu *et al.*, 2011). Additionally, a myriad of studies demonstrated a positive effect of trust on organizational loyalty (Aoki, 2020; Floh and Treiblmaier, 2015; Harris and Goode, 2004; Ribbink *et al.*, 2004).

Customer satisfaction is commonly considered a primary determinant of customer loyalty in the literature on loyalty (Deng *et al.*, 2010). Arora and Narula (2018) reviewed the literature on service quality, satisfaction and loyalty, suggesting that a chain relationship exists between service quality, satisfaction and loyalty. Therefore, we hypothesize as follows:

H5. The customer's perceived value is positively related to customer loyalty.

H6. The customer's cognitive trust toward AI chatbots positively relates to customer loyalty.

H7. The customer's affective trust toward AI chatbots is positively related to customer loyalty.

H8. Customer satisfaction with AI chatbots is positively related to customer loyalty.

4. Method

4.1 Measurement and data collection

We adopted scales that have demonstrated reliability and validity in previous studies and adapted them to this research context. To measure the items, we adopted a seven-point Likert scale. The scales of all the constructs and literature sources are listed in Table A4 in Appendix 3.

We collected survey data from a reliable data-collection platform named Wenjuanxing (the website of which is www.wjx.com). As of March 2022, this platform had more than 99.52 million users and collected 12.12 billion questionnaires from users. We required the respondents to have used at least one kind of AI chatbot and paid the platform an extra 8 RMB per questionnaire to screen for us. We asked respondents to fill out the questionnaire based on their perceptions of the most commonly used AI chatbot. After one week, 512 questionnaires were collected. We then screened the questionnaires and deleted the following three types: first, the questionnaires with the same answers to all the questions; second, the questionnaires that gave wrong answers to the attention trap questions; and third, the questionnaires that rapidly answered in a time period less than necessary. A total of 459 valid questionnaires were obtained. Patient demographics are shown in Table 2.

4.2 Measurement model test

We use SmartPLS 3.2.8 to examine the measurement and structural models. The component-based structural equation modeling (SEM) technique of SmartPLS can overcome the problem of model recognition (Reinartz *et al.*, 2009). First, we examine cross-loading, as shown in Table A5 in Appendix C. The results show that the value of items loaded on their own constructs was more than 0.7 and was significantly higher than that of the other constructs. The scale properties are summarized in Table A6 in Appendix C. Cronbach's α ranged from

Measure	Item	Frequency	Percentage (%)
Gender	Male	238	51.9
	Female	221	48.1
Age	Below 18	23	5.0
	18–25	60	13.1
	26–30	173	37.7
	31–40	120	26.1
	41–50	48	10.5
	51–60	35	7.6
Education	High school or below	62	13.5
	Three-year college	118	25.7
	Four-year college	214	46.6
	Graduate school or above	65	14.2

Table 2.
Demographics of the
research
sample (n = 459)

0.83 to 0.94. The composite reliability (CR) values of all constructs ranged from 0.90 to 0.96, which are above the threshold of 0.7, indicating good reliability of the scale. The average variance extracted (AVE) value ranged from 0.70 to 0.90, above the threshold of 0.5, indicating good convergent validity.

Regarding the discriminant validity test, the cross-loading results showed good discriminant validity. Moreover, we compare the AVE value for each variable with the squared correlation, and the former values were more significant than the latter, as shown in Table 4. The above test results suggest that the constructs have good discriminant validity.

4.3 Common method bias test

We used self-reported data to check for common method bias (CMB). Additionally, we used the marker variable technique to test CMB. As the marker variable should have no theoretical relationship with the variables in the model, we used health consciousness as a marker variable. Following Malhotra et al. (2006), we first found that the indicator of the marker variable through correlation analysis is 0.04 in this study. We then calculated the adjusted correlation and t-statistics. The results show no significant differences between the original correlations and the marker-adjusted correlations, indicating that the CMB problem was not substantial. The CMB test results are presented in Table A8 of Appendix C.

4.4 Structure model test

We calculated the structural model by running a bootstrap analysis with 5,000 resamples (Ringle et al., 2015). The test results are listed in Table 3, including the path coefficients and T-values. The total variances explained are 18.2% for perceived value, 22.4% for cognitive trust, 24.8% for affective trust, 18.8% for satisfaction and 41.6% for loyalty. We then evaluated the goodness of fit. The SRMR (standardized root mean square residual) value was 0.035, below the threshold of 0.80 (Henseler et al., 2016). Furthermore, we calculated the path coefficients between the nine first-order constructs and AI chatbot service quality. The results show that all first-order constructs significantly affected AI chatbot service quality. The VIF values of the first-order constructs are below 3.3, indicating that multicollinearity is absent (Diamantopoulos and Siguaw, 2006). The results are presented in Table 4. We then examined the partial correlations between each AI chatbot's service quality dimension and service quality. Note that we controlled for the other eight when we examined the partial correlation between one dimension and service quality. The results are presented in Table A9 in Appendix C. The results further indicate that no first-order constructs must be removed from the multidimensional framework. Examination of the effect of the control variable showed that perceived value and satisfaction significantly influenced loyalty. The influences of age, sex and education were insignificant.

Table 3.
Results of the
structural model test

Hypothesis	Path coefficient	t-values	Supported or not
H1: AI chatbot service quality → Perceived value	0.11**	2.79	Supported
H2: AI chatbot service quality → Cognitive trust	0.15***	4.01	Supported
H3: AI chatbot service quality → Affective trust	0.24***	5.43	Supported
H4: AI chatbot service quality → Satisfaction	0.36***	6.74	Supported
H5: Perceived value → Customer loyalty	0.30***	5.22	Supported
H6: Cognitive trust → Customer loyalty	0.28***	6.28	Supported
H7: Affective trust → Customer loyalty	0.57***	13.32	Supported
H8: Satisfaction → Customer loyalty	0.59***	12.57	Supported
Note(s): $p^* < 0.5$; $p^{**} < 0.01$; $p^{***} < 0.001$			

		Impact of AI service quality		
The second-order construct	The first-order constructs	Path coefficient	t-values	VIF values
AI chatbot service quality	Always available	0.18***	11.64	1.97
	Omnipresence	0.21***	12.54	1.56
	Consistency	0.20***	13.06	2.15
	Accuracy of Response	0.22***	12.75	1.68
	Availability of human service alternatives	0.19***	11.65	1.86
	Self-learning	0.20***	12.54	1.13
	Ease of use	0.14***	8.88	1.81
	Personalized recommendation	0.20***	12.60	1.88
	Human-like empathy	0.12***	6.09	2.11

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Table 4.
Results of the relationship between first- and second-order constructs

Note(s): $p^{***} < 0.001$

5. Discussion

5.1 Summary

This study verifies whether and how AI chatbot service quality influences customer loyalty. The importance of the research question lies mainly in two aspects: the importance of the research topic on service quality and customer loyalty. Second, as AI chatbots are flooding the frontline service sector, the new service context brings a new problem: the effects of AI chatbot service quality, including customer loyalty, have not been fully empirically studied. We found nine factors to present AI chatbot service quality through content analysis of the interviews and past literature review. The data examination results in the first- and second-order relationships show that the nine factors are positively and significantly related to AI chatbot service quality. The results of the structural model examination show that the internal mechanism revealed in the sequential chain model of service quality loyalty remains valid in this AI chatbot service context. AI chatbot service quality positively affects customers' perceived value, trust and satisfaction and influences customer loyalty.

5.2 Theoretical implications

First, it establishes a new research perspective to study the impact of users' perceptions of AI chatbots on organizations. As AI chatbots are gradually being popularized in frontline services, scholars have begun to study the use of AI chatbots. Existing studies regard AI chatbots as innovative information systems to study their adoption, use and continuous use, ignoring that AI chatbots are becoming service providers who directly serve customers. Consequently, previous studies examined the information system features of AI chatbots on user attitudes and behavior. From the new perspective of service providers, AI chatbots' service quality affects customers' attitudes and impressions of the organization. In addition, the present study fills the gap in the lack of empirical research on the impact of the service quality of AI chatbots on customer loyalty. Through a literature review of the existing AI chatbot research, we found that previous studies on AI chatbots include theoretical and empirical research. With regard to empirical research, the most widely researched dependent variables are consumers' attitudes and behavior toward AI chatbots, such as the adoption of AI chatbots (Sheehan *et al.*, 2020), continued use of AI chatbots (Ashfaq *et al.*, 2020; Li *et al.*, 2021), affective attachment to an AI chatbot (Lee *et al.*, 2021), consumers' trust in the chatbots (Cheng *et al.*, 2021) and perceived decision quality (Chen *et al.*, 2020). Some studies on the impact of AI chatbots on organizations have ignored research on loyalty to organizations (Crolc *et al.*, 2022; Lee *et al.*, 2021). This study validates the effect of AI chatbot service quality on customer loyalty, which enriches empirical research on AI chatbot services.

Second, it extends the sequential chain model of service quality loyalty in a new context in two ways. On the one hand, although the sequential chain model of quality-loyalty is used and validated in the literature on service quality and loyalty, it is not fully studied in the AI service context. In this new context, the service subject has changed from humans to machines, and the connotation of service quality has fundamentally changed. For example, the traditional SERVQUAL classification may not sufficiently reflect the characteristics of AI services. Therefore, whether the sequential chain model starting from service quality is valid has not been studied empirically. This study verifies the effectiveness of this framework in an AI service environment. On the other hand, this study extends this sequential chain model of quality loyalty by revealing the factors of the service quality of AI chatbots and classifying trust. The reason for differentiating cognitive and affective trust is based on the context of the AI chatbot service.

AI chatbots can be used as a tool. Consumers use AI chatbots to accomplish their tasks. In contrast, AI chatbots are human-like. Consumers perceive emotional attributes through interactions. Previous studies pointed out that consumers have two kinds of perceptions of AI chatbots: cognitive and emotional perceptions (Yam *et al.*, 2020). In our research context, cognitive and emotional perceptions were reflected in cognitive and emotional trust. The original sequential chain model of service quality loyalty did not classify trust. This study further explores the different types of trust in this model and the impact of service quality on different types of trust. Thus, this study extends the sequential chain model by revealing the impact of cognitive and emotional trust. Using the sequential chain model, this study reveals the internal mechanism of the influence of service quality on customer loyalty.

Third, this study enriches the research on service quality in the IS field. This study captures the characteristics of AI chatbots, which differ from those of traditional IS. In addition to information system-related features, this study captures the context-specific characteristics of AI chatbot services, which include accuracy of response, availability of human service alternatives, self-learning, personalized recommendations and human-like empathy. These characteristics distinguish AI chatbots from traditional self-service IS and cause the service quality dimensions of websites or IS to be no longer suitable for the AI context (Douglas *et al.*, 2003; Jiang *et al.*, 2002; Loiacono *et al.*, 2007; Tate and Evermann, 2010). In addition, because the AI chatbot is essentially an information system, the existing literature on the service quality of vendors in online contexts may not be suitable for evaluating AI chatbot service quality (Kim and Lim, 2001; Luo *et al.*, 2012). This study summarizes the attributes of AI chatbot service quality, thereby developing research on service quality in the IS field.

5.3 Practical implications

The practical implications of this study are as follows. First, the study reminds managers that they should pay attention to AI chatbots' service quality, which may influence consumer loyalty to the organization. If an AI chatbot is regarded only as an information system, the consideration of its improvement will be limited to the system quality perspective. In the customer service role, AI chatbot service quality differs from system quality. Customer use of an AI chatbot represents human-like interaction characteristics. When customers enjoy the service provided by AI chatbots, they treat it as customer service and evaluate its quality. The results showed that improving AI chatbots' service quality can enhance customer loyalty to an organization. As robots have inevitably entered frontline service jobs, the service quality of AI chatbots has become essential to promote purchasing and has far-reaching value for organizations. This can also help cultivate loyal users. Organizations can thus obtain greater value from loyal customers, such as a positive attitude toward the organization, continuous purchasing and recommendations to others. Therefore, in

addition to improving the system quality of AI chatbots by deeming them an information system, it is necessary to emphasize enhancing their service quality.

Second, this study helps organizations improve their user experience in the process of interacting with AI. In this study, AI chatbot service quality reflects customers' perceptions of AI chatbots after interacting with AI. The results also show that these service quality factors significantly affect customers' perceived value, cognitive trust, affective trust and satisfaction. These psychological states are also associated with customer loyalty. When organizations want to retain customers, they can present the high service quality of AI chatbots more visibly. Based on our research findings, the antecedents of perceived value, trust and satisfaction include nine factors of AI chatbot service quality. Organizations can display the advantages of AI chatbots regarding service quality on the chat interface. For example, as for self-learning, an AI chatbot can tell a consumer that "you are welcome to speak to me. The more we communicate, the better I will perform." As for the availability of human service alternatives, Organizations can display a reminder on the chat interface that says, "click me, you can find a human customer service immediately, and we will work together to serve you."

Third, the attributes of AI chatbots' service quality proposed in this study provide guidance through the AI chatbots' development, adoption and post-adoption stages. These attributes are critical elements that should be considered for organizations or departments that develop AI chatbots. They can also use these attributes to examine whether the AI chatbot they developed reaches a high-quality service standard. Organizations or departments that may purchase AI chatbots from development organizations can use these attributes as a reference to purchase AI chatbot services. Organizations can use these attributes to create questionnaires to investigate customers. Organizations can then use feedback to effectively improve the service quality of AI chatbots.

6. Limitations and future research

First, the type of AI chatbot investigated in this study was online virtual AI chatbots. AI chatbots include embodied types such as humanoid chatbots in banks or hospital halls. However, this study did not consider the service quality of these embodied chatbots, and future research can further explore the service quality of embodied customer service in various contexts. Second, this study relied on a cross-sectional design. To develop a causal relationship, it would be interesting to collect longitudinal data to investigate how users' attitudes and behavior change. For example, future research can investigate how customers' perceived value and trust toward an AI chatbot change in the pre- and post-purchase stages. Third, we did not test the impact of perceived value on trust because of the focus of the research aim and highlighting of the research contribution. In future research, researchers can conceptualize and classify the perceived value of chatbot services and examine the impact of perceived value on trust, satisfaction and loyalty.

7. Conclusion

This study explores whether and how customers' perception of the service quality of AI chatbots affects their loyalty to the organization. Based on the sequential chain model of service quality loyalty, this study first classifies AI chatbots' service quality into nine factors and then investigates their impact on customer loyalty through perceived value, trust and satisfaction. The results show that the sequential chain model of service quality loyalty remains valid in the context of AI services. This study enriches the research on service quality and extends the sequential chain model of service quality loyalty by capturing its features in this context. Finally, the research findings help organizations understand the key

elements for improving customer loyalty, which can help improve the service quality of AI chatbots at different stages of AI chatbot adoption.

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Study	Research question/ Research object	Research method	AI chatbot attributes/ functions	Variables
Araujo (2018)	Explores the extent to which human-like cues of chatbot influence perceptions about social presence as well as mindful and mindless anthropomorphism	Empirical	Human-like cues such as language style and name	IV: Human-like agent, intelligent agency framing DV: Mindless and mindful anthropomorphism, social presence
Ashfaq <i>et al.</i> (2020)	Investigate drivers of users' satisfaction and continuance intention toward chatbot-based customer service	Empirical	Provide 24/7 services, accessibility, flexibility, low cost, interacting with users through natural language, answers queries effectively	IV: Perceived enjoyment, information quality, service quality, perceived usefulness, perceived ease of use, need for interaction with a service employee DV: Continuance intention Mediator: Satisfaction Moderator: Need for interaction with a service employee
Bakhasi (2018)	Discuss the latest developments in the chatbot technology domain and provide readers with example applications	Conceptual	Intent recognition, dialog management, humanization, interaction channels, task automation capability, reporting and monitoring, ease of implementation, ease of implementation, security and compliance	N/A
Chen <i>et al.</i> (2020)	For each type of shopping task, which kind of decisional guidance and communication style are most accurate for ACA to help customers get the best decisional outcome?	Empirical	Immediate responses, 24/7 availability	IV: Decisional guidance, communication style DV: Perceived decision quality, perceived decision effort Mediator: Perceived cognitive fit Moderators: Shopping task, perceived task complexity
Cheng <i>et al.</i> (2021)	Explore consumers' trust and response to a text-based chatbot in e-commerce	Empirical	Empathy, friendliness	IV: Empathy, friendliness DV: Consumers' trust toward the chatbot Moderators: Task complexity and disclosure of the text-based chatbot

(continued)

Table A1.
The research on AI
chatbots

Study	Research question/ Research object	Research method	AI chatbot attributes/ functions	Variables
Choi et al., (2020)	How social perceptions regarding the warmth and competence of service robots influence consumer reactions to service failures and recovery efforts by robots?	Empirical	Deliver socially and emotionally interactive service	IV: Humanoid vs. Non-humanoid, service failure type DV: Satisfaction, behavioral intentions Mediator: Warmth Moderator: Human intervention
Crolic et al. (2022)	Examine circumstances under which anthropomorphism of customer service chatbots may be harmful for firm	Empirical	Chatbots converse with customers, through either voice or text, to address a variety of customer needs, Chabot anthropomorphism	IV: Chatbot anthropomorphism DV: Customer satisfaction, firm evaluation, purchase intentions Mediator: Increased preinteraction expectations Moderator: Customer anger N/A
Hoyer et al. (2020)	Propose a new framework for understanding the role of new technologies on the customer/shopper journey	Conceptual	(1) In pre-transaction: Selecting, advising and customizing (2) In transaction (3) Partnering and negotiating as part of the transaction (4) In post-transaction: Giving feedback and recommending additional consumption	N/A
Jenneboer et al. (2022)	Review the impact of AI chatbot on customer loyalty	Conceptual	Service quality, system quality, information quality	N/A
Lee et al., (2021)	Investigates the determinants of effective human and artificial intelligence relationship-building strategies for brands	Empirical	Based on natural language understanding, multimodal intelligence, speech recognition, problem solving, empathetic conversational systems, machine learning, IQ capacities, including efficient, accessible, relevant and updated information retrieval. EQ capacities, including empathetic conversation systems that mimic human-human communication	IV: Perceived emotional support, perceived informational support, perceived esteem support of DV: Affective attachment to an AI chatbot, purchase intention Mediators: Social attraction during human-AI interactions, emotional credibility during human-AI interactions

Table A1.

(continued)

Study	Research question/ Research object	Research method	AI chatbot attributes/ functions	Variables
Li <i>et al.</i> (2021)	What makes users continuously use chatbot services in the context of online travel agency?	Empirical	Understandability, reliability, responsiveness, assurance, interactivity	IV (Independent Variables): Understandability, reliability, responsiveness, assurance, interactivity DV (Dependent Variables): Use continuance Mediators: Confirmation, satisfaction Moderator: Technology anxiety
Luo <i>et al.</i> (2019)	What are the effects of a disclosure of chatbot identity before the machine–customer conversation in customer purchase context?	Empirical	Understand complex and subtle dialogs and address consumer requests with depth, compassion, and even humor, chatbots can converse in a friendly way with customers	IV: Disclosure before conversation DV: Purchase rate, call length Mediators: Perceived knowledge of disclosed chatbot, perceived empathy of disclosed chatbot N/A
Murtarelli <i>et al.</i> (2021)	Discuss conceptually the ethical challenges related to chatbots within the marketplace	Conceptual	Simulate human conversations, combined with predictive analytic, chatbots have the capacity to generate significant information asymmetry, provide personalized service	
Sheehan <i>et al.</i> (2020)	Investigates the relationship between miscommunication and adoption for customer service chatbots	Empirical	Anthropomorphism	IV: Error-free chatbot vs. clarification chatbot vs error chatbot DV: Adoption intent Mediator: Anthropomorphism Moderator: Need for human interaction N/A
Stieglitz <i>et al.</i> (2022)	How should conversational agents be designed to improve social media crisis communication of emergency management agencies?	Conceptual	Sociability, proactive communication, transparency, flexibility, usability	
Trivedi (2019)	Examining the customer experience of using chatbot and its impact on brand love in bank chatbots using context	Empirical	N/A	IV: Service quality, system quality, information quality DV: Brand love Moderator: Perceived risk Mediator: Customer experience

(continued)

Table A1.

Table A1.

Study	Research question/ Research object	Research method	AI chatbot attributes/ functions	Variables
Wang et al. (2021)	How do citizens' effective use of AI affect value creation in the public sector?	Empirical	Improved efficiency and faster delivery of services and possibly more rational decisions	IV: Effective use of AI voice robot service DV: Perceived public service value Moderators: Gender, experience Mediators: Perceived usefulness, perceived enjoyment, procedural justice, trust in government

Table A2.

The dimensions of service quality in the existing studies

Research context (Literature)	Dimension	Research context (Literature)	Dimension
Online shopping (Zeithaml et al., 2000)	Access, ease of navigation, efficiency, flexibility, reliability, personalization, security/privacy, responsiveness, assurance/trust, site aesthetics, price knowledge	E- quality (Madu and Madu, 2002)	Performance, features, structure, aesthetics, reliability, storage capacity, serviceability, security and system integrity, trust, responsiveness, product/service differentiation and customization, web store policies, reputation, assurance, empathy
E- Commerce (Cox and Dale, 2001) Commerce (Gefen, 2002)	Accessibility, communication, credibility, appearance Tangible, responsiveness, reliability, assurance, empathy	Online shopping (Yoo and Donthu, 2001) Web-based service (Li et al., 2002)	Ease of use, aesthetic design, processing speed, security Responsiveness, competence, quality of information, empathy, web assistance, call-back systems
Website (Barnes and Vidgen, 2002)	Usability, information, service interaction	E-service (Parasuraman et al., 2005)	Efficiency, fulfillment, system availability, privacy
Website (Cai and Jun, 2003)	Website design/content, trustworthiness, prompt/reliable service, communication	E-retailing (Wolfenbarger and Gilly, 2003)	Website design, reliability, privacy/security, customer service
Online Shopping (Ribbink et al., 2004)	Ease of use, website design, customization, responsiveness, assurance	E-service (Santos, 2003)	Incubative dimension: Ease of use, appearance, linkage, structure and layout, content Active dimension: Reliability, efficiency, support, communication, security, incentive

(continued)

Research context (Literature)	Dimension	Research context (Literature)	Dimension
Online Service (Surjadjaja <i>et al.</i> , 2003)	Service marketing: Trusted services, internal communication, external communication, price, return process Service delivery: Real time Assistance by CSR, fulfillment, availability Service design: Responsiveness, site effectiveness and functionality, up to date information, supply chain information, system integration, personalization, customization, navigability, security, interactivity, service recovery	B2C online shopping (Cenfetelli <i>et al.</i> , 2008)	Requirements: Needing, specifying Acquisition: Sourcing, ordering, paying, obtaining, Accepting Ownership: Training, monitoring, maintaining, upgrading Retirement: Accounting for, reselling/returning, replacing, evaluating Service quality: Tangibles, reliability, responsiveness, assurance, empathy
Retailing (Collier and Bienstock, 2006)	Process quality: Privacy, design, information accuracy, ease of use, functionality Outcome quality: Order timeliness, order accuracy, order condition Recovery quality: Interactive fairness, procedural fairness, outcome fairness	E-service (Fassnacht and Koese, 2006)	Environment quality: Graphic quality, clarity of layout Delivery quality: Attractiveness of selection, information quality, ease of use, technical quality Outcome quality: Reliability, functional benefit, emotional benefit
Online Shopping e-satisfaction (Evanschitzky <i>et al.</i> , 2004)	Convenience, product offerings, product information, site design, financial security	Internet retail service quality (IRSQ) (Janda <i>et al.</i> , 2002)	Performance, access, security, sensation, information
Online Shopping (Kim and Stoel, 2004)	Service quality: Reliability, responsiveness, assurance, empathy Website quality: Information quality, system quality	Online shopping for apparel (Kim <i>et al.</i> , 2006)	Efficiency, fulfillment, system availability, privacy, responsiveness, contact, personalization, information, graphic styles
Online Shopping (Semeijn <i>et al.</i> , 2005)	Assurance, navigation, e-scape, accuracy, responsiveness, customization	E-service (Xu <i>et al.</i> , 2013)	Information quality: Completeness, accuracy, format, currency, system quality reliability, flexibility, accessibility, timeliness Service quality: Tangibles, responsiveness, empathy, service reliability, assurance
Service (Zeithaml, 2002)	Information availability, ease of use or usability, privacy/security, graphic style, fulfillment	E-service (Kim and Stoel, 2004)	Web appearance, entertainment, informational fit-to-task, transaction capability, response time, trust

(continued)

Table A2.

Research context (Literature)	Dimension	Research context (Literature)	Dimension
Government service quality (Tan <i>et al.</i> , 2013)	Requirements: Needing, customizing Acquisition: Sourcing, trying, ordering, paying, tracking, accepting, authorizing Ownership: Upgrading, scheduling, delegating, negotiating, evaluating Efficient it mediated service delivery: Accessibility, navigability, interactivity, interoperability, adaptability, security	Information Systems (Petter <i>et al.</i> , 2013)	System quality: Ease of use, system flexibility, system reliability, ease of learning, intuitiveness, sophistication, flexibility, response time Information quality: Relevance, understandability, accuracy, conciseness, completeness, currency, timeliness, usability Service quality: Responsiveness, accuracy, reliability, technical competence, empathy of the personnel staff Reliability, responsiveness, rapport, tangibles
Government service quality (Connolly <i>et al.</i> , 2010)	Efficiency, system availability, fulfilment, privacy, responsiveness, compensation, contact	Information Systems (Kettinger and Lee, 2005)	
E-service failure (Tan <i>et al.</i> , 2017)	Information failures: Inaccurate information, incomplete information, irrelevant information, untimely information Functional failures: Needs recognition failure, alternatives identification failure, alternatives evaluation failure, acquisition failure, post-purchase failure System failures: System failures, inaccessibility, non-adaptability non-navigability, delay, insecurity	Firm's website (Agarwal and Venkatesh, 2002)	Content: Relevance, media use, depth/breath, Current information Ease of use: Goals, structure, feedback, promotion Made-for-the-medium: community, personalization, refinement Emotion: Challenge, plot, character strength, pace
Online shopping (Barnes and Vidgen, 2001)	Tangibles: Aesthetics, navigation Reliability: Reliability, competence Responsiveness: Responsiveness, access Assurance: Credibility, security Empathy: Communication, understanding the individual	Online Shopping (Childers <i>et al.</i> , 2001)	Navigation: Influence ease of use, enjoyment Convenience: Influence usefulness, ease of use, enjoyment Substitutability: Experience usefulness, enjoyment
E-Service (Srinivasan <i>et al.</i> , 2002)	customization, contact interactivity, care, community, convenience, cultivation, choice, character of e-retailer	Online shopping (Devaraj <i>et al.</i> , 2002)	Ease of use, usefulness, asset specificity, uncertainty, empathy, reliability, responsiveness, assurance
Website quality (Douglas <i>et al.</i> , 2003)	Presentation, content, accessibility, reliability, customer support, security	Portal sites quality (Gounaris and Dimitriadis, 2003)	Customer care, risk reduction benefit, information benefit, interaction facilitation benefit

Table A2.

(continued)

Research context (Literature)	Dimension	Research context (Literature)	Dimension
Content based websites (Gummerus <i>et al.</i> , 2004)	User interface, responsiveness, need fulfillment, security	Online shopping for apparel (Kim and Lim, 2001)	Web appearance, entertainment, information fit-to-task, transaction capability, response time, trust
Website (Loiacono <i>et al.</i> , 2007)	Information fit-to-task, tailored information, trust, response time, ease of understanding, intuitive operations, visual appeal, innovativeness, emotional appeal, consistent image, on-line completeness, relative advantage	E-retailing (Luo <i>et al.</i> , 2012)	Customer support, order tracking, on-time delivery, product met expectation, product availability, ease of finding product, site design, clarity of product info, product selection
Online shopping (McKinney <i>et al.</i> , 2002)	IQ expectations: Relevance, understandability, reliability, adequacy, scope, usefulness SQ expectations: Access, usability, entertainment, hyperlinks, navigation, interactivity	Online shopping (Melian-Alzola and Padron-Robaina, 2006)	Tangibility: Navigation, signposting, tools, explanation
Website Quality (Rosen and Purinton, 2004)	Coherence, complexity, legibility, mystery	Online Shopping (Wang, 2008)	Information quality, system quality, service quality
Website (Schubert, 2004)	Ease of use criteria, usefulness criteria, trust criteria	Airline website quality (Shchiglik and Barnes, 2004)	Web information quality, web interaction quality, web design quality
E-retailing (Shim <i>et al.</i> , 2002)	Ease of contact, customer service information, ease of access of product information	B2C E-commerce (Singh, 2002)	E-search, E-response, E-transaction, E-payment E-assurance, trust, E-help, E-technologies
Online shopping (Palmer, 2002)	Download delay: Initial access speed, speed of display between pages Navigation/organization: Arrangement, sequence, links, layout Interactivity: Customization, interactivity Responsiveness: Feedback, FAQ Information/content: amount of information, variety of information, word count, content quality	News content based website (Zhang <i>et al.</i> , 2001)	Basic: Features the support expected needs of users Performance: Features that enable the website to stay current to users' expectations Exciting: Features that are not expected but have the ability to excite and delight users

(continued)

Table A2.

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Research context (Literature)	Dimension	Research context (Literature)	Dimension
Web-based learning systems (Chiu <i>et al.</i> , 2007)	Information quality: Accuracy, completeness, ease of understanding, relevance System quality: Availability, ease of use, reliability, response time Service quality: Support	System quality, information quality and service quality (DeLone and McLean, 2003)	System quality: Adaptability, availability, reliability, response time, usability Information quality: Completeness, ease of understanding, personalization, relevance, security Service quality: Assurance, empathy, responsiveness tangibles, reliability, responsiveness, assurance, empathy
Information systems (Jiang <i>et al.</i> , 2002)	Reliability, responsiveness, assurance, empathy	Information Systems (Tate and Evermann, 2010)	

Table A2.

Appendix 2

The semi-structured interview questionnaire mentioned in Section 2 is shown as follows:

(1) What are your most commonly used AI chatbots?

(2) What’s your impression of AI chatbots service? In your opinion, what are the characteristics of AI chatbots?

(3) Are you satisfied with the service provided by the AI chatbots? Why?

(4) Do you feel dissatisfied with the AI chatbots’ service in some aspects? Why?

Table A3.

The examples of interview samples, open codes and axial codes

Examples of interview samples	Open codes	Axial codes
A1. AI chatbots respond rapidly and are always available A2. AI chatbots are always online and never leave work A3. AI chatbots are 24/7 available	Always available, always online, never leave work, 24/7 available	Always available
B1. I can use the AI chatbot anywhere. I feel it is very convenient B2. Using an AI chatbot is not limited by location B3. The AI chatbot can be used through various devices. Using it on a mobile device is convenient	Use the AI chatbot anywhere, not limited by location, mobility	Omnipresence
C1. The service quality of the AI chatbot is very stable C2. The AI chatbot will not feel tired as human employees, and the AI chatbot is emotionally stable C3. The information provided by the AI chatbot is consistent	Emotional stable, information is consistent	Consistency

(continued)

Examples of interview samples	Open codes	Axial codes
D1. The AI chatbot sometimes cannot understand the questions I ask D2. The answer from the AI chatbot is standardized. I wonder if the chatbot understands the question I asked and cannot give a specific answer D3. The AI chatbot cannot understand complex problems and cannot understand the subtext of my response	Understand the questions I ask, understand complex problems, understand the subtext of my response	Accuracy of response
E1. When encountering problems that cannot be solved, the AI chatbot transfers them to human customer service to solve E2. AI chatbot can collaborate with human customer service without consumers' repeating the problem E3. It is convenient to switch from AI chatbot service to manual customer service E4. On some websites, it is difficult to find human customer service. When the AI chatbot cannot solve the problem, I don't know how to do the next step	AI chatbot transfers problems to human customer service, collaborate with human customer service, switch from AI chatbot service to manual customer service, difficult to find human customer service	Availability of human service alternatives
F1. I find that the AI chatbot is getting to know me more and more, and it keeps track of my needs F2. The quality of AI chatbot's answers is continuously improving F3. AI chatbot has self-learning ability. It can understand users better after multiple interactions	Getting to know me more and more, keep track of consumers' needs, quality of answers is continuously improving, has self-learning ability, understand users better after multiple interactions	Self-learning
G1. The AI chatbot answers questions clearly G2. The AI chatbot shows me brief functional guidance. I think interacting with the AI chatbot is very simple G3. You can ask the AI chatbot questions without typing too many words, and it can give you an answer. It is very easy to use	Answer questions clearly, interacting with the AI chatbot is very simple, easy to use	Ease of use
H1. The AI chatbot can make accurate recommendations based on my preferences H2. I am very satisfied with the accurate recommendation of the AI chatbot. The recommended products fit my needs H3. The AI chatbots' recommendations are intelligent, saving my shopping search time	Make accurate recommendations, recommended products fit consumers' needs, recommendations are intelligent	Personalized recommendation
I1. The answers of the AI chatbots are standardized and have no emotions I2. The AI chatbot can't perceive my emotions and can't understand my anxiety I3. I feel the AI chatbot is very concerned about my needs	Have no emotions, can't perceive consumers' emotions, concerned about consumers' needs	Empathy

Table A3.

Appendix 3

Constructs	Indicators	Literature source
Always available	(1) This AI chatbot is available at 7 days × 24 h (2) I can get the AI chatbot service at any time (3) This AI chatbot is always online (4) I can always find the AI chatbot online (5) This AI chatbot will respond to me at any time (6) I can get this AI chatbot at any time throughout the day	Venkatesh <i>et al.</i> (2011)
Omnipresence	(1) I can contact this AI chatbot anywhere (2) I can use the AI chatbot on multiple devices (3) Using the AI chatbot will not be limited by place	Baek and Yoo (2018)
Consistency	(1) The quality of service provided by this AI chatbot is stable (2) This AI chatbot always tries to provide me with a relatively high level of service (3) There is little difference in the quality of each service provided by the AI chatbot (4) This AI chatbot will not have many fluctuations in quality (5) The service quality of this AI chatbot will not change from time to time	Alge (2001)
Accuracy of response	(1) This AI chatbot can accurately understand what I say (2) The response from this AI chatbot is accurate (3) The answer of this AI chatbot corresponds to the question I asked (4) The answer from this AI chatbot meets my expectations (5) This AI chatbot can answer my questions accurately	Mayer and Davis (1999)
Availability of human service alternatives	(1) Human service is readily available. For example, when the AI chatbot cannot solve the problem, I can easily find a human customer service (2) I do not encounter difficulties in obtaining human service alternative (3) I can easily obtain human service when necessary to continue the service	Furneaux and Wade (2011)
Self-learning	(1) This AI chatbot can learn from past experience (2) This AI chatbot can become better through learning (3) This AI chatbot is enhanced through learning (4) This AI chatbot can learn to improve themselves well	Rijsdijk <i>et al.</i> (2007)
Ease of use	(1) It is easy to become skillful at using the AI chatbot (2) Learning to interact with the AI chatbot is easy (3) The AI chatbot is flexible to interact with (4) My interaction with the AI chatbot is clear and understandable	Gefen <i>et al.</i> (2003)
Personalized recommendation	(1) In general, most items recommended by this AI chatbot match my preferences very well (2) In general, most items recommended by this AI chatbot fit my tastes very well (3) In general, most items recommended by this AI chatbot are interesting to me	Zhang <i>et al.</i> (2011)

Table A4.
The scale of
measurement

(continued)

Constructs	Indicators	Literature source
Human-like empathy	(1) I feel that this AI chatbot puts my interests first (2) I feel that this AI chatbot is serving me attentively (3) This AI chatbot makes me feel concerned	Pitt <i>et al.</i> (1995)
Cognitive trust	(1) This AI chatbot works with professionalism and dedication (2) Through the use experience of this AI chatbot, I see no reason to doubt the AI chatbot's competence and preparation for a job (3) I believe using this AI chatbot would not make my problems solving more complex	Stewart and Gosain (2006)
Affective trust	(1) I feel secure about relying on this AI chatbot for my decision (2) I feel comfortable about relying on this AI chatbot for my decision (3) I feel content about relying on this AI chatbot for my decision	Komiak and Benbasat (2006)
Perceived value	(1) This AI chatbot is an excellent value for my time and effort (2) I am very happy to spend time interacting with this AI chatbot (3) I think it is valuable for me to spend time and effort interacting with this AI chatbot (4) Interacting with this AI chatbot is worth my time and effort	Harris and Goode (2004)
Satisfaction	(1) Overall, I feel extremely satisfied with this AI chatbot (2) Overall, I feel extremely pleased with this AI chatbot (3) My expectations of this AI chatbot is achieved (4) I would recommend this AI chatbot to a friend	Fang <i>et al.</i> (2014)
Customer loyalty	(1) I will suggest this organizations' products or services to other customers (2) I intend to keep purchasing products or service from this organization (3) I will recommend this organizations' products or service to others	Fang (2019)

Note(s): Please choose one of the chatbots that you use the most, recall your interaction experience with it, and answer the following questions

Table A4.

Table A5.
Cross loading results

	ACC	AT	AVA	AVAH	CONS	CT	EASE	EMP	LOY	OMNI	PER	PV	SAT	SEL
ACC1	0.89	0.22	0.22	0.22	0.24	0.10	0.21	0.21	0.36	0.55	0.23	0.34	0.40	0.27
ACC2	0.92	0.22	0.23	0.22	0.21	0.12	0.16	0.22	0.35	0.56	0.20	0.34	0.40	0.28
ACC3	0.93	0.21	0.23	0.22	0.21	0.13	0.14	0.22	0.34	0.56	0.16	0.35	0.39	0.27
ACC4	0.86	0.14	0.18	0.12	0.19	0.09	0.14	0.15	0.35	0.47	0.16	0.30	0.38	0.25
ACC5	0.85	0.15	0.22	0.16	0.24	0.10	0.19	0.22	0.33	0.47	0.17	0.31	0.40	0.20
AT1	0.21	0.91	0.11	0.14	0.14	0.10	0.15	0.08	0.33	0.22	0.18	0.28	0.35	0.14
AT2	0.19	0.95	0.17	0.16	0.14	0.14	0.17	0.08	0.33	0.21	0.20	0.34	0.39	0.17
AT3	0.19	0.89	0.20	0.15	0.19	0.16	0.17	0.14	0.30	0.17	0.21	0.30	0.36	0.15
AVAI	0.25	0.15	0.87	0.47	0.23	0.15	0.18	0.52	0.29	0.19	0.22	0.28	0.28	0.09
AVA2	0.20	0.15	0.84	0.49	0.20	0.11	0.09	0.56	0.32	0.09	0.17	0.30	0.24	0.10
AVA3	0.23	0.12	0.86	0.54	0.27	0.19	0.18	0.56	0.31	0.15	0.22	0.31	0.26	0.12
AVA4	0.20	0.16	0.90	0.54	0.23	0.19	0.16	0.59	0.31	0.13	0.19	0.29	0.24	0.10
AVA5	0.17	0.15	0.84	0.49	0.22	0.14	0.15	0.57	0.31	0.12	0.19	0.31	0.25	0.05
AVA6	0.20	0.16	0.87	0.51	0.26	0.13	0.14	0.59	0.28	0.11	0.21	0.29	0.25	0.09
AVAH1	0.23	0.13	0.52	0.89	0.17	0.25	0.15	0.59	0.24	0.18	0.16	0.32	0.29	0.11
AVAH2	0.19	0.12	0.50	0.87	0.17	0.18	0.15	0.55	0.25	0.16	0.21	0.33	0.29	0.09
AVAH3	0.15	0.16	0.52	0.87	0.15	0.23	0.10	0.54	0.21	0.13	0.11	0.28	0.20	0.08
CONS1	0.25	0.15	0.28	0.16	0.86	0.09	0.55	0.20	0.36	0.14	0.55	0.33	0.33	0.19
CONS2	0.19	0.13	0.24	0.17	0.82	0.10	0.51	0.19	0.31	0.14	0.54	0.32	0.27	0.16
CONS3	0.21	0.15	0.24	0.17	0.86	0.11	0.53	0.19	0.30	0.13	0.57	0.32	0.29	0.20
CONS4	0.20	0.16	0.20	0.15	0.86	0.11	0.53	0.16	0.31	0.11	0.53	0.29	0.28	0.14
CONS5	0.20	0.13	0.21	0.15	0.84	0.11	0.55	0.16	0.32	0.16	0.54	0.33	0.25	0.17
CT1	0.10	0.13	0.14	0.25	0.11	0.93	0.16	0.22	0.30	0.10	0.15	0.29	0.24	0.15
CT2	0.12	0.12	0.18	0.23	0.12	0.96	0.14	0.23	0.30	0.14	0.16	0.26	0.22	0.14
CT3	0.12	0.15	0.18	0.24	0.11	0.95	0.12	0.23	0.32	0.15	0.13	0.26	0.26	0.15
EASE1	0.15	0.13	0.14	0.11	0.53	0.09	0.82	0.10	0.24	0.10	0.45	0.20	0.27	0.16
EASE2	0.19	0.18	0.15	0.14	0.56	0.15	0.88	0.13	0.30	0.13	0.51	0.30	0.28	0.18
EASE3	0.16	0.16	0.16	0.13	0.54	0.15	0.94	0.13	0.28	0.15	0.51	0.27	0.27	0.18
EASE4	0.15	0.17	0.17	0.13	0.59	0.14	0.88	0.09	0.28	0.12	0.54	0.28	0.27	0.18
EMP1	0.18	0.07	0.55	0.53	0.18	0.22	0.11	0.86	0.27	0.08	0.11	0.26	0.21	0.09
EMP2	0.26	0.10	0.57	0.55	0.19	0.19	0.11	0.86	0.30	0.15	0.17	0.31	0.29	0.09
EMP3	0.14	0.12	0.58	0.56	0.18	0.21	0.11	0.87	0.29	0.08	0.16	0.26	0.24	0.05

(continued)

	ACC	AT	AVA	AVAH	CONS	CT	EASE	EMP	LOY	OMNI	PER	PV	SAT	SEL
LOY1	0.34	0.31	0.32	0.25	0.32	0.28	0.29	0.29	0.91	0.30	0.30	0.51	0.53	0.33
LOY2	0.31	0.30	0.31	0.25	0.31	0.28	0.27	0.30	0.82	0.27	0.30	0.43	0.46	0.33
LOY3	0.36	0.31	0.29	0.19	0.35	0.28	0.25	0.27	0.87	0.32	0.33	0.43	0.51	0.31
OMNI1	0.50	0.21	0.14	0.18	0.17	0.15	0.14	0.13	0.35	0.93	0.14	0.30	0.36	0.23
OMNI2	0.58	0.20	0.14	0.17	0.16	0.15	0.14	0.12	0.32	0.96	0.15	0.29	0.33	0.22
OMNI3	0.58	0.20	0.14	0.16	0.12	0.10	0.12	0.10	0.29	0.91	0.13	0.25	0.29	0.21
PER1	0.17	0.22	0.20	0.15	0.57	0.14	0.54	0.15	0.30	0.14	0.88	0.25	0.30	0.20
PER2	0.17	0.20	0.24	0.17	0.58	0.12	0.50	0.17	0.31	0.11	0.87	0.26	0.30	0.18
PER3	0.20	0.14	0.17	0.14	0.54	0.15	0.46	0.15	0.33	0.15	0.87	0.28	0.31	0.16
PV1	0.32	0.31	0.32	0.32	0.36	0.25	0.26	0.31	0.48	0.26	0.25	0.88	0.47	0.31
PV2	0.28	0.25	0.28	0.29	0.28	0.24	0.21	0.27	0.41	0.22	0.23	0.85	0.48	0.30
PV3	0.36	0.31	0.28	0.31	0.32	0.24	0.27	0.25	0.43	0.33	0.27	0.81	0.49	0.35
PV4	0.29	0.28	0.29	0.27	0.31	0.24	0.27	0.27	0.47	0.21	0.27	0.87	0.49	0.31
SAT1	0.39	0.35	0.27	0.27	0.25	0.20	0.20	0.25	0.55	0.31	0.26	0.50	0.85	0.40
SAT2	0.36	0.32	0.24	0.24	0.28	0.19	0.27	0.25	0.45	0.29	0.30	0.46	0.86	0.34
SAT3	0.32	0.31	0.20	0.20	0.26	0.25	0.24	0.23	0.44	0.30	0.25	0.47	0.80	0.31
SAT4	0.39	0.36	0.27	0.26	0.32	0.22	0.32	0.24	0.48	0.28	0.34	0.46	0.83	0.41
SEL1	0.26	0.15	0.13	0.09	0.14	0.19	0.17	0.09	0.32	0.25	0.15	0.31	0.37	0.89
SEL2	0.25	0.16	0.10	0.06	0.21	0.11	0.19	0.06	0.34	0.18	0.21	0.33	0.37	0.87
SEL3	0.27	0.14	0.06	0.10	0.17	0.14	0.16	0.07	0.34	0.22	0.17	0.33	0.44	0.92
SEL4	0.24	0.14	0.09	0.13	0.18	0.11	0.18	0.10	0.33	0.18	0.20	0.35	0.40	0.88

Note(s): ACC: Accuracy of response; AT: Affective trust; AVA: Always available; AVAH: Availability of human service alternatives; CONS: Consistency; CT: Cognitive trust; EASE: Ease of use; EMP: Human-like empathy; LOY: Customer loyalty; LOY: Customer loyalty; OMNI: Omnipresence; PER: Personalized recommendation; PV: Perceived value; SAT: Satisfaction; SEL: Self-learning

Table A5.

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Constructs	Indicators	Factor loading	Cronbach's α	CR	AVE
Accuracy of response	ACC1	0.89	0.94	0.95	0.79
	ACC2	0.92			
	ACC3	0.93			
	ACC4	0.86			
	ACC5	0.85			
Affective trust	AT1	0.91	0.90	0.94	0.84
	AT2	0.95			
	AT3	0.89			
Always availability	AVA1	0.87	0.93	0.95	0.74
	AVA2	0.84			
	AVA3	0.86			
	AVA4	0.90			
	AVA5	0.84			
	AVA6	0.87			
Availability of human service alternatives	AVAH1	0.89	0.85	0.91	0.77
	AVAH2	0.87			
	AVAH3	0.87			
Consistency	CONS1	0.86	0.90	0.93	0.72
	CONS2	0.82			
	CONS3	0.86			
	CONS4	0.86			
	CONS5	0.84			
Cognitive trust	CT1	0.93	0.94	0.96	0.90
	CT2	0.96			
	CT3	0.95			
Ease of use	EASE1	0.82	0.90	0.93	0.78
	EASE2	0.88			
	EASE3	0.94			
	EASE4	0.88			
Human-like empathy	EMP1	0.86	0.83	0.90	0.74
	EMP2	0.86			
	EMP3	0.87			
Customer loyalty	LOY1	0.91	0.84	0.90	0.75
	LOY2	0.82			
	LOY3	0.87			
Omnipresence	OMNI1	0.93	0.93	0.95	0.87
	OMNI2	0.96			
	OMNI3	0.91			
Personalized recommendation	PER1	0.88	0.85	0.91	0.76
	PER2	0.87			
	PER3	0.87			
Perceived value	PV1	0.88	0.88	0.91	0.73
	PV2	0.85			
	PV3	0.81			
	PV4	0.87			
Satisfaction	SAT1	0.85	0.86	0.90	0.70
	SAT2	0.86			
	SAT3	0.80			
	SAT4	0.83			
Self-learning	SEL1	0.89	0.91	0.94	0.79
	SEL2	0.87			
	SEL3	0.92			
	SEL4	0.88			

Table A6.
Scale properties

	ACC	AT	AVA	AVAH	CONS	CT	EASE	EMP	LOY	OMNI	PER	PV	SAT	SEL
ACC	0.89													
AT	0.41	0.92												
AVA	0.24	0.37	0.86											
AVAH	0.21	0.36	0.59	0.88										
CONS	0.25	0.37	0.27	0.18	0.85									
CT	0.32	0.34	0.38	0.45	0.32	0.95								
EASE	0.19	0.38	0.17	0.15	0.63	0.35	0.88							
EMP	0.23	0.31	0.66	0.64	0.21	0.44	0.13	0.86						
LOY	0.39	0.55	0.35	0.27	0.38	0.32	0.31	0.33	0.87					
OMNI	0.59	0.42	0.15	0.18	0.16	0.34	0.14	0.13	0.34	0.93				
PER	0.20	0.41	0.23	0.17	0.64	0.36	0.57	0.18	0.36	0.15	0.87			
PV	0.37	0.54	0.34	0.35	0.37	0.48	0.30	0.32	0.53	0.30	0.30	0.85		
SAT	0.44	0.40	0.29	0.29	0.33	0.46	0.31	0.29	0.58	0.35	0.35	0.57	0.84	
SEL	0.29	0.37	0.11	0.11	0.20	0.35	0.20	0.09	0.37	0.23	0.21	0.37	0.44	0.89
Note(s): ACC: Accuracy of response; AT: Affective trust; AVA: Always available; AVAH: Availability of human service alternatives; CONS: Consistency; CT: Cognitive trust; EASE: Ease of use; EMP: Human-like empathy; LOY: Customer loyalty; OMNI: Omnipresence; PER: Personalized recommendation; PV: Perceived value; SAT: Satisfaction; SEL: Self-learning														

Table A7.
The comparison
between AVE values
and the squared
correlation

Table A8.
The result of CMB test

[illegible]

	ACC	AT	AVA	AVAH	CONS	CT	EASE	EMP	LOY	OMNI	PER	PV	SAT
<i>Panel C: t-test statistics of marker-adjusted correlation matrix</i>													
AT	8.52												
AVA	4.55	7.82											
AVAH	3.84	7.55	14.93										
CONS	4.79	7.82	5.27	3.15									
CT	6.51	7.03	8.09	10.09	6.51								
EASE	3.38	8.09	2.92	2.46	16.64	7.29							
EMP	4.31	6.26	18.06	17.10	3.84	9.79	2.01						
LOY	8.36	13.39	7.29	5.27	8.09	12.33	6.26	6.77					
OMNI	14.93	9.20	2.46	3.15	2.69	7.03	2.24	2.01	6.73				
PER	3.61	8.92	4.31	2.92	17.10	7.55	14.14	3.15	7.23	2.37			
PV	7.82	13.03	7.03	7.29	7.82	11.01	6.01	6.51	12.03	7.23	5.76		
SAT	9.79	8.64	5.76	5.76	6.77	10.39	6.26	5.76	13.73	12.03	6.98	13.38	
SEL	5.76	2.92	2.56	2.56	3.61	2.46	3.61	2.11	7.48	13.73	3.69	7.48	9.34
Note(s): ACC: Accuracy of response; AT: Affective trust; AVA: Always available; AVAH: Availability of human service alternatives; CONS: Consistency; CT: Cognitive trust; EASE: Ease of use; EMP: Human-like empathy; LOY: Customer loyalty; OMNI: Omnipresence; PER: Personalized recommendation; PV: Perceived value; SAT: Satisfaction; SEL: Self-learning													

Table A8.

Table A9.
The partial correlation
between the first-order
constructs and the
second order construct

Correlations	Control variables								
	ACC	AVA	AVAH	CONS	EASE	EMP	OMNI	PER	SEL
Partial correlation ACC-SQ		0.62***	0.61***	0.63***	0.65***	0.61***	0.46***	0.64***	0.57***
Correlation ACC-SQ		0.62***	0.62***	0.62***	0.62***	0.62***	0.62***	0.62***	0.62***
Partial correlation AVA-SQ	0.65***		0.48***	0.65***	0.17***	0.45***	0.67***	0.23***	0.69***
Correlation AVA-SQ	0.65***		0.65***	0.65***	0.65***	0.65***	0.65***	0.65***	0.65***
Partial correlation AVAH-SQ	0.55***	0.29***		0.60***	0.59***	0.31***	0.55***	0.58***	0.58***
Correlation AVAH-SQ	0.56***	0.56***		0.56***	0.56***	0.56***	0.56***	0.56***	0.56***
Partial correlation CONS-SQ	0.70***	0.69***	0.71***		0.49***	0.70***	0.71***	0.47***	0.69***
Correlation CONS-SQ	0.68***	0.68***	0.68***		0.68***	0.68***	0.68***	0.68***	0.68***
Partial correlation EASE-SQ	0.55***	0.24***	0.32***	0.58***		0.61***	0.58***	0.59***	0.60***
Correlation EASE-SQ	0.56***	0.56***	0.56***	0.56***		0.56***	0.56***	0.56***	0.56***
Partial correlation EMP-SQ	0.64***	0.66***	0.64***	0.31***	0.65***		0.63***	0.39***	0.60***
Correlation EMP-SQ	0.61***	0.61***	0.61***	0.61***	0.61***		0.61***	0.61***	0.61***
Partial correlation OMNI-SQ	0.21***	0.53***	0.49***	0.54***	0.53***	0.53***		0.53***	0.46***
Correlation OMNI-SQ	0.50***	0.50***	0.50***	0.50***	0.50***	0.50***		0.50***	0.50***
Partial correlation PER-SQ	0.65***	0.64***	0.64***	0.33***	0.42***	0.65***	0.64***		0.61***
Correlation PER-SQ	0.62***	0.62***	0.62***	0.62***	0.62***	0.62***	0.62***		0.62***
Partial correlation SEL-SQ	0.43***	0.57***	0.53***	0.51***	0.49***	0.55***	0.46***	0.48***	
Correlation SEL-SQ	0.50***	0.50***	0.50***	0.50***	0.50***	0.50***	0.50***	0.50***	

Note(s): $p^{***} < 0.001$

ACC: Accuracy of response; AVA: Always available; AVAH: Availability of human service alternatives; CONS: Consistency; EASE: Ease of use; EMP: Human-like empathy; OMNI: Omnipresence; PER: Personalized recommendation; SEL: Self-learning; SQ: Service quality

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Impact of AI
service quality

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